MASTERING DECIMALS

EXERCISE 3A: Complete Solutions & Explanations

This reference guide provides comprehensive, step-by-step solutions for **Exercise 3A** from the textbook *'Mastering Decimals: Theory, Operations, and Applications'*. Every solution is meticulously worked out, highlighting the underlying mathematical principles (such as finding like decimals, expanding numbers, and converting between fractions and decimals) as detailed in the course material. This document serves as an excellent study resource and a verification key for your exercises.

Question 1: Convert the following decimals into like decimals

UNDERLYING CONCEPT: LIKE DECIMALS

Like decimals are decimals having the same number of decimal places. To convert unlike decimals into like decimals, we:
1. Identify the maximum number of decimal places (say, *n*) among the given decimals.
2. Convert each of the other decimals into one having *n* decimal places by annexing zeros to the extreme right of the decimal part. Annexing zeros does not change its value (e.g., 0.6 = 0.60).

(i) 8.1, 9.4, 6.05, 12
- Maximum decimal places: **2** (in 6.05).
- Convert each to 2 decimal places:
  - 8.1 = **8.10**
  - 9.4 = **9.40**
  - 6.05 = **6.05**
  - 12 = **12.00**
- **Like decimals:** 8.10, 9.40, 6.05, 12.00

(ii) 6.35, 11.8, 9.125, 32.04, 9
- Maximum decimal places: **3** (in 9.125).
- Convert each to 3 decimal places:
  - 6.35 = **6.350**
  - 11.8 = **11.800**
  - 9.125 = **9.125**
  - 32.04 = **32.040**
  - 9 = **9.000**
- **Like decimals:** 6.350, 11.800, 9.125, 32.040, 9.000

(iii) 6.005, 2, 0.6, 1.78, 14.3
- Maximum decimal places: **3** (in 6.005).
- Convert each to 3 decimal places:
  - 6.005 = **6.005**
  - 2 = **2.000**
  - 0.6 = **0.600**
  - 1.78 = **1.780**
  - 14.3 = **14.300**
- **Like decimals:** 6.005, 2.000, 0.600, 1.780, 14.300

(iv) 0.72, 53.76, 16, 8.304, 2.3754
- Maximum decimal places: **4** (in 2.3754).
- Convert each to 4 decimal places:
  - 0.72 = **0.7200**
  - 53.76 = **53.7600**
  - 16 = **16.0000**
  - 8.304 = **8.3040**
  - 2.3754 = **2.3754**
- **Like decimals:** 0.7200, 53.7600, 16.0000, 8.3040, 2.3754

Question 2: Write each of the following decimals in expanded form

UNDERLYING CONCEPT: EXPANDED FORM

Expanded form expresses a decimal as the sum of its individual place values. These include whole number places (ones, tens, hundreds) and fractional decimal places (tenths = 1/10, hundredths = 1/100, thousandths = 1/1000, etc.). We provide both fractional expansion and decimal expansion formats.

(i) 16.74

- Place Values: 1 Ten, 6 Ones, 7 Tenths, 4 Hundredths
- **Fractional Expansion:**
  10 + 6 + 7/10 + 4/100
- **Product Form:**
  (1 × 10) + (6 × 1) + (7 × 1/10) + (4 × 1/100)
- **Decimal Expansion:**
  10 + 6 + 0.7 + 0.04

(ii) 243.689

- Place Values: 2 Hundreds, 4 Tens, 3 Ones, 6 Tenths, 8 Hundredths, 9 Thousandths
- **Fractional Expansion:**
  200 + 40 + 3 + 6/10 + 8/100 + 9/1000
- **Product Form:**
  (2 × 100) + (4 × 10) + (3 × 1) + (6 × 1/10) + (8 × 1/100) + (9 × 1/1000)
- **Decimal Expansion:**
  200 + 40 + 3 + 0.6 + 0.08 + 0.009

(iii) 9.3578

- Place Values: 9 Ones, 3 Tenths, 5 Hundredths, 7 Thousandths, 8 Ten-thousandths
- **Fractional Expansion:**
  9 + 3/10 + 5/100 + 7/1000 + 8/10000
- **Product Form:**
  (9 × 1) + (3 × 1/10) + (5 × 1/100) + (7 × 1/1000) + (8 × 1/10000)
- **Decimal Expansion:**
  9 + 0.3 + 0.05 + 0.007 + 0.0008

(iv) 2314.57

- Place Values: 2 Thousands, 3 Hundreds, 1 Ten, 4 Ones, 5 Tenths, 7 Hundredths
- **Fractional Expansion:**
  2000 + 300 + 10 + 4 + 5/10 + 7/100
- **Product Form:**
  (2 × 1000) + (3 × 100) + (1 × 10) + (4 × 1) + (5 × 1/10) + (7 × 1/100)
- **Decimal Expansion:**
  2000 + 300 + 10 + 4 + 0.5 + 0.07

Question 3: Express each of the following as a fraction in simplest form

UNDERLYING CONCEPT: CONVERTING DECIMALS TO FRACTIONS

To express a decimal as a fraction in its simplest form:
1. Write the decimal without the decimal point as the numerator of the fraction.
2. Write 1 followed by as many zeros as there are decimal places in the given decimal as the denominator.
3. Simplify the fraction by dividing both numerator and denominator by their Greatest Common Divisor (GCD).

(i) 0.8
• Fraction: 8/10
• Divide numerator & denominator by 2:
(8 ÷ 2) / (10 ÷ 2) = **4/5**

(ii) 0.45
• Fraction: 45/100
• Divide numerator & denominator by 5:
(45 ÷ 5) / (100 ÷ 5) = **9/20**

(iii) 1.6
• Fraction: 16/10
• Divide numerator & denominator by 2:
(16 ÷ 2) / (10 ÷ 2) = **8/5** (or **1 3/5**)

(iv) 4.75
• Fraction: 475/100
• Divide numerator & denominator by 25:
(475 ÷ 25) / (100 ÷ 25) = **19/4** (or **4 3/4**)

(v) 0.345
• Fraction: 345/1000
• Divide numerator & denominator by 5:
(345 ÷ 5) / (1000 ÷ 5) = **69/200**

(vi) 64.015
• Fraction: 64015/1000
• Divide numerator & denominator by 5:
(64015 ÷ 5) / (1000 ÷ 5) = **12803/200** (or **64 3/200**)

(vii) 12.725
• Fraction: 12725/1000
• Divide numerator & denominator by 25:
(12725 ÷ 25) / (1000 ÷ 25) = **509/40** (or **12 33/40**)

(viii) 0.0024
• Fraction: 24/10000
• Divide numerator & denominator by 8:
(24 ÷ 8) / (10000 ÷ 8) = **3/1250**

Question 4: Convert each of the following fractions into a decimal

UNDERLYING CONCEPT: CONVERTING FRACTIONS TO DECIMALS

Type 1: Denominator is 10 or a power of 10
• Shift the decimal point of the numerator to the left by as many places as there are zeros in the denominator.
Type 2: Other Denominators
• Convert the fraction into an equivalent fraction with a denominator of 10, 100, 1000, etc., by multiplying both the numerator and the denominator by an appropriate number, then apply the Type 1 rule. Alternatively, perform long division of the numerator by the denominator.

(i) 19/10
• Denominator is 10 (1 zero).
• Shift decimal 1 place left: 19 → **1.9**

(ii) 243/100
• Denominator is 100 (2 zeros).
• Shift decimal 2 places left: 243 → **2.43**

(iii) 51/100
• Denominator is 100 (2 zeros).
• Shift decimal 2 places left: 51 → **0.51**

(iv) 9/100
• Denominator is 100 (2 zeros).
• Shift decimal 2 places left: 9 → **0.09**

(v) 72/1000
• Denominator is 1000 (3 zeros).
• Shift decimal 3 places left: 72 → **0.072**

(vi) 3/5
• Multiply numerator & denominator by 2 to make denominator 10:
(3 × 2) / (5 × 2) = 6/10 = **0.6**

(vii) 7/8
• Multiply numerator & denominator by 125 to make denominator 1000:
(7 × 125) / (8 × 125) = 875/1000 = **0.875**

(viii) 37/40
• Multiply numerator & denominator by 25 to make denominator 1000:
(37 × 25) / (40 × 25) = 925/1000 = **0.925**

(ix) 4 3/8
• Convert to sum: 4 + 3/8
• Multiply fractional part by 125:
(3 × 125) / (8 × 125) = 375/1000 = 0.375
• Add whole part: 4 + 0.375 = **4.375**

(x) 9 1/25
• Convert to sum: 9 + 1/25
• Multiply fractional part by 4 to make denominator 100:
(1 × 4) / (25 × 4) = 4/100 = 0.04
• Add whole part: 9 + 0.04 = **9.04**

(xi) 2 9/16
• Convert to sum: 2 + 9/16
• Multiply fractional part by 625 to make denominator 10000:
(9 × 625) / (16 × 625) = 5625/10000 = 0.5625
• Add whole part: 2 + 0.5625 = **2.5625**

(xii) 4 11/125
• Convert to sum: 4 + 11/125
• Multiply fractional part by 8 to make denominator 1000:
(11 × 8) / (125 × 8) = 88/1000 = 0.088
• Add whole part: 4 + 0.088 = **4.088**